Lab1 Submission

Remote Team Time-Tracking & Project Approver

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Section: G

Problem Statement #46 | Developer Tools & IT Operations

Deliverable 1:

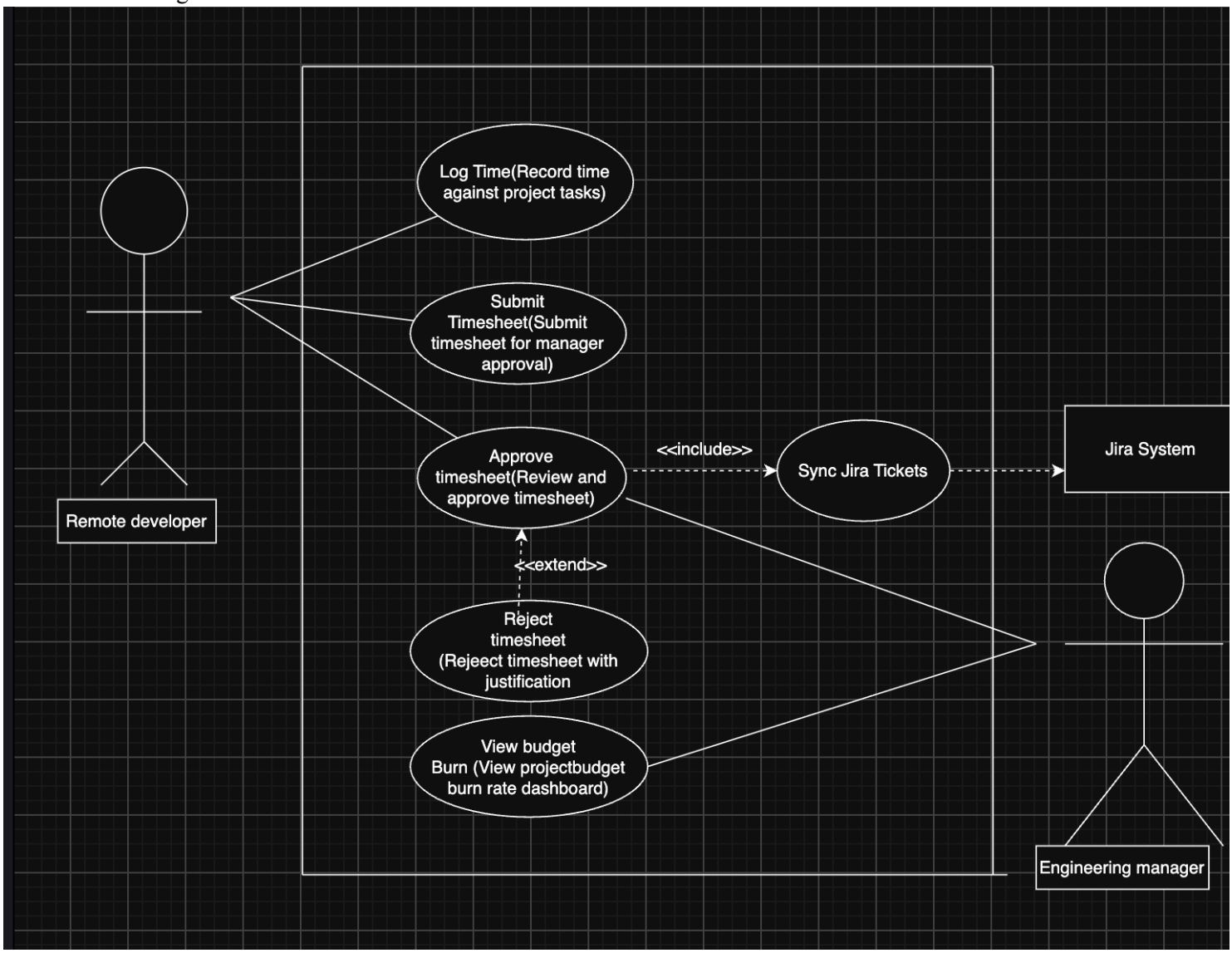
Requirements Table

Requirements derived from the provided Remote Team Time-Tracking & Project Approver scenario (Problem Statement #46).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow developers to log time against specific project tasks and prevent timesheet submissions exceeding 24 hours in a single calendar day.	High	Pass: Valid timesheet submitted for manager approval. Fail: Timesheet logging 30 hours in one day accepted.	Given by problem statement; core data-capture capability the whole system depends on.
FR-002	Functional	The system shall allow a Remote Developer to submit a completed weekly timesheet to their assigned Engineering Manager for approval.	High	Pass: A submitted timesheet appears in the manager's pending-approval queue with status "Submitted". Fail: A timesheet with unsaved/incomplete daily entries can be submitted.	Formalizes the handoff between logging time and manager review, which the approval workflow depends on.
FR-003	Functional	The system shall allow an Engineering Manager to approve or reject a submitted timesheet, requiring a comment when rejecting.	High	Pass: Rejecting a timesheet without entering a comment is blocked; approving updates status to "Approved". Fail: A timesheet is rejected with no comment recorded.	Core managerial control point required by the problem statement's approval workflow.
FR-004	Functional	The system shall automatically sync approved logged hours to their corresponding Jira tickets, updating ticket time-spent fields.	Medium	Pass: After approval, the linked Jira ticket's logged-time field reflects the approved hours within one sync cycle. Fail: Approved hours are not reflected on the Jira ticket.	Keeps engineering effort data consistent between the tracker and the system of record (Jira), avoiding duplicate manual entry.
FR-005	Functional	The system shall calculate and display each project's real-time budget burn rate as approved hours multiplied by role billing rate, aggregated per project.	High	Pass: Displayed burn rate equals the manually computed sum of (approved hours x billing rate) for the project, within $0.01. Fail: Displayed burn rate differs from the manual calculation by more than $0.01.	Core value proposition for Engineering Managers: visibility into project budget consumption.
NFR -001	Nonfunctional - Performance & Security	Timesheet approval status transitions and project budget burn charts must update across dashboards within 500 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Dashboard update latency exceeds 500 ms, or security benchmarks are not met.	Given by problem statement; ensures managers see accurate, timely budget and approval information.
NFR -002	Nonfunctional - Reliability & Auditability	The system shall retain an immutable audit log of every timesheet submission, approval, and rejection for a minimum of 3 years, encrypted at rest.	Medium	Pass: Automated audit report reconstructs the full approval history for any timesheet up to 3 years old; log entries are confirmed encrypted at rest. Fail: A historical approval action cannot be reconstructed, or log entries are found unencrypted.	Supports payroll disputes, compliance, and manager accountability for approval decisions.

Deliverable 2:

UML use case diagram



Deliverable 2:

Use-Case Flow: Approve Timesheet

Primary Actor: Engineering Manager

Related Use Cases: includes Sync Jira Tickets; extended by Reject Timesheet.

Preconditions

- The Engineering Manager is authenticated and logged into the application.
- At least one timesheet has been submitted by a Remote Developer and is awaiting approval.

Postconditions

- On success: the timesheet status is updated to “Approved,” approved hours are synced to the linked Jira tickets, and the project budget burn rate is recalculated.
- On failure/rejection: the timesheet status is updated to “Rejected” with the manager's comment attached, and it stays excluded from the burn-rate calculation until resubmitted.

Main Success Scenario

1. The Engineering Manager opens the Pending Timesheets list and selects a submitted timesheet.
2. The system shows the logged hours per task, cross-checked against the linked Jira tickets.
3. The Engineering Manager reviews the entries and taps “Approve.”
4. The system updates the timesheet status to “Approved” and syncs the approved hours to the corresponding Jira tickets.
5. The system recalculates the project's budget burn rate and refreshes the dashboard within 500 ms.
6. The system notifies the Remote Developer that their timesheet was approved.
7. The use case ends successfully.

Alternate Flow – 3a: Manager Rejects Timesheet

3a1. Instead of approving, the Engineering Manager taps “Reject” and must enter a comment explaining why.

3a2. The system sets the timesheet status to “Rejected” and stores the manager's comment against it.

3a3. The system notifies the Remote Developer of the rejection and the reason; the developer can edit the logged hours and resubmit, sending it back to the pending-approval queue.